

[28] Spurious but systematic correlation in functional connectivity MRI networks arise from subject motion

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Executive Academic Summary & Connectomic Relevance

Demonstrates how micro-head motion artificially inflates short-range correlations and decreases long-range correlations in rs-fMRI, establishing Framewise Displacement (FD) censoring.

Mathematical & Neurobiological Foundations

• $\text{FD}_i = |\Delta d_{ix}| + |\Delta d_{iy}| + |\Delta d_{iz}| + |\Delta \alpha_i| + |\Delta \beta_i| + |\Delta \gamma_i|$

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